

ClaimGuard: AI-Powered Insurance Fraud Detection

Executive Business Case - For CFO / Board Approval

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1. PROBLEM STATEMENT

Kenya's motor insurance sector loses an estimated KES 40-60 billion annually to fraudulent claims (IRA Kenya, 2024). The current manual review process takes 5-15 days per claim and catches only 40% of fraud. Each missed fraudulent claim costs an average of KES 150,000. For a mid-size insurer processing 2,000 claims per month, this is KES 14.4M in undetected fraud losses every month - a direct hit to the combined ratio and shareholder returns.

The core failure is structural: human reviewers cannot cross-reference hundreds of claims simultaneously. A garage filing nine fraudulent claims across different policyholders over 90 days is completely invisible to an adjuster reading one file at a time.

2. PROPOSED SOLUTION

ClaimGuard is a production-ready AI claims intelligence platform with three detection layers:

- * Rules Engine: 8 hard motor fraud rules (claim within 14 days of inception, duplicate OB numbers, provider billing spikes, round-sum amounts, missing documents)
- * XGBoost ML Scorer: 25 engineered features -> fraud probability 0-1. AUC-ROC 0.84 | Precision 79% | Recall 81%
- * Network / Ring Detector: Graph analysis across providers, claimants, and vehicles - identifies organised fraud rings invisible to individual reviewers

Every decision includes a plain-English SHAP explanation for adjusters and a full audit trail for IRA compliance. Processing time: under 30 seconds per claim.

3. FINANCIAL IMPACT

Assumptions: 2,000 claims/month | 8% fraud rate = 160 fraudulent claims | Avg value KES 150,000

Source	Calculation	Monthly	Annual
Additional fraud caught	$(81\%-40\%) \times 160 \times \text{KES } 150\text{K}$	KES 9.84M	KES 118.1M
Adjuster time savings	$95\% \text{ reduction} \times 8 \text{ staff} \times \text{KES } 120\text{K}$	KES 0.91M	KES 10.9M
False positive reduction	$40\% \text{ fewer wrongly flagged} \times \text{KES } 5\text{K}$	KES 1.60M	KES 19.2M
TOTAL SAVINGS		KES 12.35M	KES 148.2M

Investment: KES 11.5M (Year 1) - Development KES 8.5M | Cloud KES 1.8M | Maintenance KES 1.2M

1,188% Year 1 ROI	28 Days Payback Period	KES 580M 5-Year NPV
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4. RISKS & MITIGATION

Risk: Regulatory challenge - IRA questions automated rejections

Mitigation: Every rejection includes a full SHAP audit trail. Human review is mandatory for all rejections. Designed to comply with IRA data governance guidelines.

Risk: Model drift - fraud patterns evolve faster than the model retrains

Mitigation: Automated monthly retraining with champion/challenger model registry. Alert fires if AUC-ROC drops below 0.78.

5. RECOMMENDATION - GO v

Approve full deployment of ClaimGuard.

KES 11.5M investment. KES 148.2M Year 1 return. 28-day payback. The technology is production-ready.

Approvals required: (1) Budget KES 11.5M (2) IT window Q4 2026 (3) IRA compliance sign-off